

THE 20:26 PROJECT

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Project Title: Explainable Temporal Graph Neural Networks for Cryptocurrency Fraud Detection

Project Summary (Problem & Motivation)

Cryptocurrency networks such as Bitcoin and Ethereum operate as decentralized transaction graphs, where users interact through peer-to-peer transfers. While this transparency is powerful, it also enables fraudulent activities such as money laundering, Ponzi schemes, and illicit fund flows. Traditional machine learning approaches struggle to capture both the relational structure (who transacts with whom) and the temporal dynamics (when transactions occur), which are critical for identifying suspicious patterns.

Graph Neural Networks (GNNs) have emerged as a powerful tool for modeling such relational data, but most existing approaches either ignore temporal evolution or lack interpretability. This is a major limitation in fraud detection, where understanding *why* a transaction is flagged is as important as detecting it.

This project aims to develop an **Explainable Temporal Graph Neural Network (TGNN)** framework that models both the structural and temporal aspects of blockchain transactions while providing interpretable insights into predictions. The goal is to improve fraud detection accuracy while making the model transparent enough for real-world financial and regulatory use. This is particularly interesting because it sits at the intersection of **AI, finance, and cybersecurity**, with direct real-world impact.

Approach

The project will involve implementing a **Temporal Graph Neural Network pipeline** for node/edge classification (fraud vs. non-fraud transactions). Initially, a baseline model such as a standard GNN (e.g., GCN or GraphSAGE) will be implemented using libraries like **PyTorch Geometric (PyG)** or **DGL** to establish performance benchmarks.

Next, temporal dynamics will be incorporated using architectures such as **Temporal Graph Networks (TGN)** or **Dynamic Graph Neural Networks**, which capture evolving transaction patterns over time. Features such as transaction amount, frequency, timestamps, and wallet connectivity will be encoded into node and edge representations.

To go beyond “just running models,” we will:

- Experiment with **different temporal encoding strategies** (time embeddings, snapshot-based graphs, continuous-time models).
- Compare **static vs. temporal GNN performance** on fraud detection tasks.
- Implement **attention mechanisms** (e.g., GAT-style layers) to identify influential neighbors.

For explainability, we will integrate techniques like **GNNExplainer** or **PGExplainer** to identify subgraphs and features that contribute most to fraud predictions. Analyze how explanations differ between fraudulent and normal transactions, providing insights into behavioral patterns.

Finally, we will evaluate the models using metrics such as **F1-score, ROC-AUC, and precision-recall curves**, and perform ablation studies to understand the impact of temporal and structural components.

Resources / Related Work & Papers

Recent research in graph-based fraud detection has shown promising results, particularly with models like **Graph Convolutional Networks (GCN)** and **Graph Attention Networks (GAT)** for transaction classification. However, these models typically operate on static graphs and fail to capture the temporal evolution of blockchain networks.

Temporal models such as **Temporal Graph Networks (TGN)** and **DySAT (Dynamic Self-Attention Network)** extend GNNs by incorporating time-aware embeddings, allowing the model to learn evolving patterns in transaction flows. These approaches are particularly relevant for fraud detection, where malicious behavior often unfolds over time.

Explainability in GNNs is an active research area, with methods like **GNNExplainer** and **PGExplainer** providing ways to interpret predictions by identifying important subgraphs and features. Combining temporal modeling with explainability remains relatively underexplored, making this project both relevant and novel.

Recent work has also emphasized the importance of dataset quality and curation in improving model performance. For instance, Gadre et al. (2023) introduce **DataComp**, a large-scale benchmarking framework for evaluating dataset construction strategies in multimodal learning. Their work highlights that improvements in dataset design can significantly impact downstream model performance, sometimes even more than architectural changes. This reinforces the importance of selecting and preprocessing high-quality transaction datasets for fraud detection tasks.

Key papers include:

- *Towards the Explainability of Temporal Graph Neural Networks via Motif Discovery* (Gadre et al)
- *Inductive Representation Learning on Large Graphs* (GraphSAGE)
- *GNNExplainer: Generating Explanations for Graph Neural Networks*
- *DySAT: Deep Neural Representation Learning on Dynamic Graphs*
- *Temporal Graph Networks for Deep Learning on Dynamic Graphs* (Rossi et al.)

This project builds on these works by integrating **temporal learning + explainability** specifically for **fraud detection in crypto networks**, which is still an emerging area.

Datasets (with Links)

1. Elliptic Bitcoin Dataset (Best Choice)

- <https://www.kaggle.com/datasets/ellipticco/elliptic-data-set>
- Contains **real Bitcoin transactions** labeled as **licit/illicit**
- Includes **temporal features + graph structure**
- Industry-standard for crypto fraud detection research

2. Ethereum Transaction Dataset (Optional Extension)

- <https://www.kaggle.com/datasets/bigquery/ethereum-blockchain>
- Large-scale dataset for experimenting with scalability
- Requires preprocessing to construct graphs

3. IEEE / AMLSim Synthetic Dataset (Advanced)

- <https://github.com/IBM/AMLSim>
- Simulated anti-money laundering dataset
- Useful for testing model robustness

4. DataComp Benchmark (NeurIPS 2023 – Reference Dataset Framework)

- https://proceedings.neurips.cc/paper_files/paper/2023/hash/5c5bc3553815adb4d1a8a5b8701e41a9-Abstract-Conference.html
- Provides a **large-scale benchmark (12.8B image-text pairs)** for studying dataset quality and filtering strategies

- Useful as a **reference framework for dataset curation and evaluation**, even though it is not specific to cryptocurrency

CITATIONS

Core Temporal Graph Neural Networks

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